

Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline

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Further medication

See the

visual summary

for an overview of the recommendations and additional information to support medicine choice up to the point at which a person starts insulin-based treatment.

Recommendations in this section that cover dipeptidyl peptidase-4 (DPP-4) inhibitors, glucagonlike peptide-1 (GLP-1) receptor agonists, sulfonylureas and sodium–glucose cotransporter-2 (SGLT-2) inhibitors refer to each of these groups of medicines at class level.

For GLP-1 receptor agonists, at the time of publication (February 2026) this only includes liraglutide, dulaglutide, and semaglutide. For subcutaneous semaglutide (Ozempic), this only includes doses up to 1 mg once a week.

SGLT-2 inhibitors and GLP-1 receptor agonists are recommended as much for their cardiovascular and renal benefits as for their glycaemic benefits (unless otherwise specified).

Healthcare professionals should also refer to the summary of product characteristics for individual medicines for contraindications and precautions to take in pregnancy and breastfeeding and for women, trans men and non-binary people of childbearing potential.

1.25 People with type 2 diabetes and no relevant comorbidity

1.25.1

For adults with type 2 diabetes and

no relevant comorbidity

who need further medicines to reach their individualised

glycaemic targets

offer to add a DPP-4 inhibitor to their current treatment

if this is contraindicated, not tolerated or is not effective, offer to add:

a sulfonylurea or

pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

Why the committee made this recommendation

Most of the evidence used an additive strategy (where additional treatment was provided) rather than a switching

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strategy (where 1 treatment was stopped and another was started). This appeared to provide effective results. The committee agreed that this is likely to be effective for most people.

DPP-4 inhibitors

Evidence was available for individual population groups: people with heart failure, chronic kidney disease and at higher risk of developing cardiovascular disease. The evidence for all groups showed similar results; that DPP-4 inhibitors were effective at reducing HbA1c and relatively safe. Based on the evidence, and their own experience, the committee recommended that a DPP-4 inhibitor should be the first choice for people who need further medicines. DPP-4 inhibitors all have similar clinical efficacy, and no particular medicine was more cost effective than the others.

Sulfonylureas or pioglitazone

If a DPP-4 inhibitor is not effective or tolerated, a sulfonylurea or pioglitazone should be considered. The evidence showed that both reduced HbA1c. However, there was also limited evidence, indicating that they both had a potential for adverse effects:

sulfonylureas might increase hypoglycaemic events and weight gain

pioglitazone might worsen cardiovascular outcomes and should be avoided for people with heart failure.

The committee acknowledged that pioglitazone increases the risk of fractures and weight gain, as stated in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

Insulin-based treatments

Evidence showed that insulin-based treatments are less effective than most other antidiabetic therapies at reducing harm from adverse events because it can increase the risk of hypoglycaemic events and weight gain. There are 2 main scenarios when insulin is an effective treatment:

managing acute hyperglycaemia

managing long-term worsening hyperglycaemia that does not respond to other treatments.

However, in the committee's experience, insulin may also be a good option for people who cannot tolerate more effective medicines. Therefore, they recommended insulin alongside other options for people who cannot tolerate DPP-4 inhibitors.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendation might affect practice

These recommendations are likely to be a change from current practice, with wider access to therapies with more recent evidence of clinical and cost effectiveness.

The recommendations on DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. DPP-4 inhibitors vary in price, but the default assumption is that services will use the medicine with the lowest acquisition cost (in the absence of patient-specific factors).

1.26 People with heart failure (any ejection fraction unless specified)

1.26.1

For adults with type 2 diabetes and heart failure who need further medicines to reach their individualised glycaemic targets:

offer to add a DPP-4 inhibitor to their current treatment

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if this is contraindicated, not tolerated or not effective, offer to add:

a sulfonylurea or

an insulin-based treatment (see the section on insulin-based treatments

[2026]

Why the committee made this recommendation

DPP-4 inhibitors

Evidence was available for individual population groups: people with heart failure, chronic kidney disease and at higher risk of developing cardiovascular disease. The evidence for all groups showed similar results, that DPP-4 inhibitors were effective at reducing HbA1c and relatively safe. Based on this evidence, and their own clinical and lived experience, the committee recommended that a DPP-4 inhibitor should be the first choice for people with heart failure who need further medicines. DPP-4 inhibitors all have similar clinical efficacy, and no particular medicine was more cost effective than the others.

Sulfonylureas

Sulfonylureas are recommended for people who need further treatment because the evidence showed that they reduced HbA1c based on the evidence for people at high risk of cardiovascular disease. However, there was also limited evidence indicating that they had a potential for adverse effects, given that sulfonylureas and insulin-based therapies might increase hypoglycaemic events and weight gain. Healthcare professionals are advised to consider this when choosing treatments.

Insulin-based treatments

Evidence showed that insulin-based treatments are less effective than most other antidiabetic therapies at reducing harm from adverse events because it can increase the risk of hypoglycaemic events and weight gain. There are 2 main scenarios for which insulin is an effective treatment:

managing acute hyperglycaemia

managing long-term worsening hyperglycaemia that does not respond to other treatments.

However, in the committee's experience, insulin may also be a good option for people who cannot tolerate other medicines. Therefore, they recommended insulin alongside sulfonylureas for people who need further medicines to reach their individualised glycaemic targets.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

These recommendations are likely to be a change from current practice, with wider access to therapies with more recent evidence of clinical and cost effectiveness.

Sulfonylureas and insulin

The recommendations on sulfonylureas and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. The use of more intensive earlier treatment may lead to a reduction in the use of insulin, which may offset costs in the long term.

1.27 People with atherosclerotic cardiovascular disease

1.27.1

If an adult with type 2 diabetes develops

atherosclerotic cardiovascular disease

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after starting initial treatment, offer to add subcutaneous semaglutide (Ozempic), up to 1 mg once a week, to their current treatment, for its cardiovascular and renal benefits.

[2026]

1.27.2

For adults with type 2 diabetes and atherosclerotic cardiovascular disease who need further medicines to reach their individualised

glycaemic targets

, offer to add to their current treatment:

a sulfonylurea or

pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

Why the committee made these recommendations

Subcutaneous semaglutide

Subcutaneous semaglutide (Ozempic), up to 1 mg once a week, was recommended for people who develop atherosclerotic cardiovascular disease after starting initial treatment because evidence showed that this medicine:

reduces the risk of cardiovascular events

helps with weight loss as a side effect

is cost effective.

The evidence showed subcutaneous semaglutide (Ozempic), up to 1 mg once a week, was the most cost effective and clinically effective GLP-1 receptor agonist in terms of cardiovascular and renal protection, and weight reduction.

Sulfonylureas and pioglitazone

Sulfonylureas and pioglitazone are recommended for people who need further treatment because the evidence showed that these both reduced HbA1c. However, there was also limited evidence indicating that they both had a potential for adverse effects:

sulfonylureas and insulin-based therapies might increase hypoglycaemic events and weight gain

pioglitazone might worsen cardiovascular outcomes and should be avoided for people with heart failure.

The committee acknowledged that pioglitazone increases the risk of fractures and weight gain, as stated in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

Insulin-based treatments

Evidence showed that insulin-based treatments are less effective than most other antidiabetic therapies at reducing harm from adverse events because it can increase the risk of hypoglycaemic events and weight gain. There are 2 main scenarios in which insulin is an effective treatment:

managing acute hyperglycaemia

managing long-term worsening hyperglycaemia that does not respond to other treatments.

However, in the committee's experience, insulin may also be a good option for people who cannot tolerate other medicines. Therefore, they recommended insulin alongside other options for people who need further medicines to reach their individualised glycaemic targets.

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Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

These recommendations are likely to be a change from current practice, with wider access to therapies with more recent evidence of clinical and cost effectiveness.

Subcutaneous semaglutide

Subcutaneous semaglutide is a GLP-1 receptor agonist. GLP-1 receptor agonists were previously recommended after triple therapy with metformin and 2 other oral medicines was not effective, tolerated or contraindicated. Therefore, recommending the treatment as triple therapy after taking metformin and 1 oral medicine (an SGLT-2 inhibitor) may lead to increases in costs. The committee agreed this will likely reduce over time, because the cardiovascular benefits and weight loss side effect of subcutaneous semaglutide will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease. An increase in subcutaneous semaglutide use will reduce DPP-4 inhibitor use. Early intervention could lead to weight loss as a beneficial side effect and so to a better long-term prognosis, which, if maintained, will reduce:

long-term treatment requirements and

the need for later stage treatments (such as insulin) and

downstream treatment costs.

Sulfonylureas, pioglitazone and insulin

The recommendations on sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. The use of more intensive earlier treatment may lead to a reduction in the use of insulin, which may offset costs in the long term.

1.28 People with early onset type 2 diabetes

1.28.1

For adults with early onset type 2 diabetes who need further medicines to reach their individualised glycaemic targets

, consider adding a GLP-1 receptor agonist or tirzepatide.

[2026]

1.28.2

For adults with early onset type 2 diabetes who need further medicines to reach their individualised glycaemic targets and for whom a GLP-1 receptor agonist or tirzepatide is contraindicated, not tolerated or not appropriate:

offer to add a DPP-4 inhibitor to their current treatment

if this is contraindicated, not tolerated or is not effective, offer to add:

a sulfonylurea or

pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

1.28.3

For adults with early onset type 2 diabetes who need further medicines to reach their individualised glycaemic targets and are taking a GLP-1 receptor agonist or tirzepatide, offer to add to their current treatment:

a sulfonylurea or

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pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

Why the committee made these recommendations

GLP-1 receptor agonists and tirzepatide

Though there were no clinical trials focusing solely on people with early onset diabetes, there was good evidence in the wider population of people with type 2 diabetes, which the committee agreed could be extrapolated to people with early onset type 2 diabetes. It showed that GLP-1 receptor agonists reduce the risk of cardiovascular events and lead to weight loss as a side effect. Based on their experience, the committee agreed that people with early onset type 2 diabetes:

have a very high lifetime risk of cardiovascular and renal complications, and of dying from them, and

are more likely to be living with obesity.

Taking this into account, they agreed that, if not started as initial therapy, starting a GLP-1 receptor agonist or tirzepatide as subsequent therapy was likely the most appropriate treatment because it could help to reduce long-term cardiovascular, renal and glycaemic complications better than other medications, while also reducing weight as a side effect.

DPP-4 inhibitors (for people not taking a GLP-1 receptor agonist or tirzepatide)

Evidence was not available for people with early onset type 2 diabetes. The evidence for groups who did not have early onset type 2 diabetes, including people at higher risk of cardiovascular disease, showed that DPP-4 inhibitors were effective at reducing HbA1c and relatively safe. Based on this evidence, and their experience, the committee recommended that a DPP-4 inhibitor should be a choice for people with early onset type 2 diabetes who need further medicines when a GLP-1 receptor agonist or tirzepatide is not appropriate. DPP-4 inhibitors all have similar clinical efficacy, and no particular medicine was more cost effective than the others.

Sulfonylureas and pioglitazone

The committee recommended sulfonylureas and pioglitazone in this group based on their own experience, and by extrapolating the evidence from other groups covered in this guideline (which showed that these medicines reduced HbA1c). However, there was also limited evidence for these other groups that showed a potential for adverse effects:

sulfonylureas might increase hypoglycaemic events and weight gain

pioglitazone might worsen cardiovascular outcomes and should be avoided for people with heart failure.

The committee acknowledged that pioglitazone increases the risk of fractures and weight gain, as stated in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

Insulin-based treatments

For insulin-based treatments, the evidence for other populations showed that they are less effective than most other antidiabetic therapies at reducing harm from adverse events because it can increase the risk of hypoglycaemic events and weight gain. There are 2 main scenarios when insulin is an effective treatment:

managing acute hyperglycaemia

managing long-term worsening hyperglycaemia that does not respond to other treatments.

However, in the committee's experience, insulin may also be a good option for people who cannot tolerate other

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medicines. Therefore, they recommended insulin alongside other options for people who need further medicines to reach their individualised glycaemic targets.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

These recommendations are likely to be a change from current practice, enabling wider access to therapies supported by recent evidence of clinical and cost effectiveness.

GLP-1 receptor agonists or tirzepatide

GLP-1 receptor agonists and tirzepatide were previously recommended after triple therapy with metformin and 2 other oral medicines was not effective, tolerated or contraindicated. Therefore, recommending the treatment as triple therapy after taking metformin and 1 oral medicine (an SGLT-2 inhibitor) may lead to increases in costs. The committee agreed that this will likely reduce over time, because the cardiovascular benefits and weight loss side effect of GLP-1 receptor agonists will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease. Increased use of GLP-1 receptor agonists or tirzepatide will reduce DPP-4 inhibitor use. Early intervention could lead to weight loss as a beneficial side effect and so to a better long-term prognosis, which, if maintained, will reduce:

long-term treatment requirements and

the need for later stage treatments (such as insulin).

DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin

The recommendations on DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. There may be cost savings if people are no longer prescribed GLP-1 receptor agonists and DPP-4 inhibitors together. DPP-4 inhibitors vary in price, but the default assumption is that services will use the medicine with the lowest acquisition cost (in the absence of patient-specific factors). The use of more intensive earlier treatment may lead to a reduction in the use of insulin, which may offset costs in the long term.

1.29 People living with obesity

1.29.1

If considering medicines primarily for weight management, see

information about medicines for overweight and obesity in NICE's guideline on overweight and obesity management [2026]

1.29.2

Consider adding a GLP-1 receptor agonist or tirzepatide for adults with type 2 diabetes who are living with obesity, if: they have been taking initial therapy for at least 3 months and further medicines are needed to reach their individualised glycaemic targets

they are not already taking a GLP-1 receptor agonist or tirzepatide.

For a definition of obesity, see the

section on classifying overweight and obesity in adults in NICE's guideline on overweight and obesity management [2026]

1.29.3

For adults with type 2 diabetes who are living with obesity who need further medicines to reach their individualised glycaemic targets and for whom a GLP-1 receptor agonist or tirzepatide is contraindicated, not tolerated, not appropriate or not effective:

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offer to add a DPP-4 inhibitor to their current treatment

if this is contraindicated, not tolerated or not effective, offer to add:

a sulfonylurea or

pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

1.29.4

For adults with type 2 diabetes who are living with obesity who need further medicines to reach their individualised glycaemic targets and are already taking a GLP-1 receptor agonist or tirzepatide, offer to add:

a sulfonylurea or

pioglitazone or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

Why the committee made these recommendations

GLP-1 receptor agonists or tirzepatide

The committee agreed GLP-1 receptor agonists and tirzepatide should be considered for people living with obesity because of the medicines' glycaemic reduction properties. GLP-1 receptor agonists also have the cardiovascular and renal benefits. These may also reduce weight, which may be an important side effect for the person with type 2 diabetes and their healthcare professionals. However, if weight reduction is the primary aim of the treatment, the committee agreed that guidance should be sought within NICE's guideline on overweight and obesity management instead of this guideline.

Evidence showed that out of the GLP-1 receptor agonists, only liraglutide reached the cost-effectiveness threshold in the base-case analysis for people living with obesity. The committee agreed that while liraglutide was cost effective, it was less clinically effective than other GLP-1 receptor agonists, for example semaglutide. Semaglutide was cost effective when more favourable assumptions around weight loss, such as it being maintained for a longer period, were used in the economic model. Given this, the committee agreed that it was plausible for both liraglutide and semaglutide to be cost effective. The committee therefore made the recommendation to consider more clinically effective GLP-1 receptor agonists dependent on the needs of the person.

Tirzepatide is a glucose-dependent insulinotropic polypeptide and glucagon-like peptide-1 dual-receptor agonist. The committee looked at the evidence from

NICE's technology appraisal guidance for tirzepatide for people with type 2 diabetes

, which recommends tirzepatide as an alternative to GLP-1 receptor agonists. In October 2023, the technology appraisal guidance recommended that it should be offered alongside diet and exercise to adults for whom type 2 diabetes is insufficiently controlled and only if triple therapy with metformin and 2 other oral antidiabetic medicines is ineffective, not tolerated or contraindicated (among other criteria). Given that the evidence supporting the technology appraisal guidance found tirzepatide was clinically and cost effective for reducing HbA1c and weight for people with type 2 diabetes, the committee for this guideline agreed to recommend it as an alternative to GLP-1 receptor agonists.

A GLP-1 receptor agonist or tirzepatide should only be added after at least 3 months of initial therapy, so that the effect of the initial therapy on the person's glycaemic targets can be assessed and taken into account when deciding whether additional treatment is needed. The committee agreed that the medicine should be continued as long as it has benefits

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in reducing HbA1c for the person.

Sulfonylureas and pioglitazone

The committee recommended sulfonylureas and pioglitazone in this group based on their own experience, and by extrapolating the evidence from other groups covered in this guideline (which showed that these medicines reduced HbA1c). However, there was also limited evidence for these other groups that showed a potential for adverse effects:

sulfonylureas and insulin-based therapies might increase hypoglycaemic events and weight gain

pioglitazone might worsen cardiovascular outcomes and should be avoided for people with heart failure.

The committee acknowledged that pioglitazone increases the risk of fractures and weight gain, as stated in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

Insulin-based treatments

Evidence on insulin-based treatments for other populations showed that these treatments are less effective than most other antidiabetic therapies at reducing harm from adverse events because it can increase the risk of hypoglycaemic events and weight gain. There are 2 main scenarios when insulin is an effective treatment:

managing acute hyperglycaemia

managing long-term worsening hyperglycaemia that does not respond to other treatments.

However, in the committee's experience insulin may also be a good option for people who cannot tolerate other medicines. Therefore, they recommended insulin alongside other options for people who need further medicines to reach their individualised glycaemic targets.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

These recommendations are likely to be a change from current practice, with wider access to therapies with more recent evidence of clinical and cost effectiveness.

GLP-1 receptor agonists and tirzepatide

GLP-1 receptor agonists and tirzepatide were previously recommended after triple therapy with metformin and 2 other oral medicines was not effective, tolerated or contraindicated. Therefore, recommending the treatment as triple therapy after taking metformin and 1 oral medicine (an SGLT-2 inhibitor) may lead to increases in costs. The committee agreed this will likely reduce over time, because the cardiovascular benefits and the weight loss side effect of GLP-1 receptor agonists will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease. An increase in GLP-1 receptor agonist or tirzepatide use will mean that DPP-4 inhibitor use will reduce. Early intervention could lead to weight loss as a beneficial side effect and so to a better long-term prognosis, which, if maintained, will reduce:

long-term treatment requirements and

the need for later stage treatments (such as insulin).

DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin

The recommendations on DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. There may be cost savings if people are no longer prescribed GLP-1 receptor agonists or tirzepatide and DPP-4 inhibitors together. DPP-4 inhibitors do vary in price, but the default assumption is that services will use the medicine with the lowest acquisition cost (in the absence of patient-specific

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factors). The use of more intensive earlier treatment may lead to a reduction in the use of insulin, which may offset costs in the long term.

1.30 People with chronic kidney disease

Medicines vary in their contraindications and precautions for use in people with renal impairment. See NICE's information on prescribing medicines and refer to the summary of product characteristics for individual products.

1.30.1

For adults with type 2 diabetes and chronic kidney disease who need further medicines to reach their individualised glycaemic targets

consider adding a DPP-4 inhibitor

if they are already taking a DPP-4 inhibitor or if a DPP-4 inhibitor is contraindicated, not tolerated or not effective, consider adding:

pioglitazone or

a sulfonylurea (if their eGFR is above 30 ml/min/1.73 m

) or

an insulin-based treatment.

[2026]

Why the committee made this recommendation

The committee made a recommendation for this group based on their knowledge and experience, because the clinical evidence was very limited.

DPP-4 inhibitors are effective at reducing HbA1c and have fewer adverse effects than other comparable options. If DPP-4 inhibitors are contraindicated, not tolerated or not effective, then in the committee's experience the best option is either pioglitazone, a sulfonylurea, or an insulin-based treatment.

The committee acknowledged that sulfonylureas increase the risk of hypoglycaemia, pioglitazone increases the risk of heart failure, fractures and weight gain, and insulin increases the risk of hypoglycaemic events and weight gain, as recorded in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendation might affect practice

The recommendations on DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. There may be cost savings if people are no longer prescribed GLP-1 receptor agonists or tirzepatide and DPP-4 inhibitors together. DPP-4 inhibitors do vary in price, but the default assumption is that services will use the medicine with the lowest acquisition cost (in the absence of patient-specific factors).

1.31 People with frailty

1.31.1

For adults with frailty who need further medicines to manage their hyperglycaemia symptoms and reach their individualised

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glycaemic targets

consider adding a DPP-4 inhibitor to their current treatment or

if they are already taking a DPP-4 inhibitor or if a DPP-4 inhibitor is contraindicated, not tolerated or is not effective, consider adding 1 of the following to their current treatment:

pioglitazone or

a sulfonylurea or

an insulin-based treatment (see the

section on insulin-based treatments

[2026]

1.31.2

When choosing a treatment with the person, take into account that sulfonylureas and insulin-based treatments can increase the risk of hypoglycaemia and falls.

[2026]

Why the committee made these recommendations

There was no evidence on outcomes for people with frailty. Using their knowledge from clinical practice, the committee recommended that, in this group, the aim of treatment should primarily be to control symptoms.

If initial therapy does not achieve the treatment goals, a DPP-4 inhibitor can reduce HbA1c with limited adverse effects.

Pioglitazone, sulfonylureas and insulin-based treatments are recommended as alternatives based on the committee's experience. The committee did not recommend one treatment over the other because of the lack of evidence and the diverse needs of people with frailty.

The committee acknowledged that sulfonylureas and insulin increase the risk of hypoglycaemia and falls, while pioglitazone increases the risk of fractures and weight gain, as recorded in the BNF. As with each medicine listed in this section, they agreed that healthcare professionals should consider the benefits and risks of each treatment when deciding if a treatment is appropriate, on a case-by-case basis. This information should be discussed with the person with diabetes so that they and their healthcare professional can come to an informed decision together about their treatment plan.

The committee did not recommend GLP-1 receptor agonists or tirzepatide for people with frailty. However, they agreed that there is no inherent safety risk for this population. Therefore, if a person has a relevant indication and frailty, they can still be offered a GLP-1 receptor agonist or tirzepatide.

Because of the lack of evidence, the committee made a

recommendation for research on treatment strategies for people with type 2 diabetes and frailty

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

The recommendations on DPP-4 inhibitors, sulfonylureas, pioglitazone and insulin do not reflect a significant change in current practice and are unlikely to increase resource use. There may be cost savings if people are no longer prescribed GLP-1 receptor agonists or tirzepatide and DPP-4 inhibitors together. DPP-4 inhibitors do vary in price, but the default assumption is that services will use the medicine with the lowest acquisition cost (in the absence of patient-specific factors).